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LOW TEMPERATURE THERMODYNAMIC PROPERTIES OF MARTIAN SHERGOTTITES. C. P. Opeil,[1,2] G. J. Consolmagno,[1] D. T. Britt[3], and R. J. Macke[1] [1]Vatican Observatory (V-00120, Vatican City State, brother_guy@me.com; macke@jesuits.net) [2]Department of Physics, Boston College (Chestnut Hill, MA 02467, USA, opeil@bc.edu) [3]Center for Lunar and Asteroid Surface Science, and Department of Physics, University of Central Florida (Orlando, FL 32816, USA, dbritt@ucf.edu).

Introduction: Shergottites are the most abundant type of Martian meteorite arriving on Earth. Petrological analyses of Martian meteorites indicate that Mars is dominated by igneous rocks with basaltic compositions and products of limited fractional crystallization. [1] Meteorites from Mars are recognized by the composition of trapped Martian atmospheric gas in impact-melted glasses, distinctive bulk oxygen isotopic compositions, and Fe/Mn ratios in pyroxenes. [2]

We report low-temperature specific heat capacity, and thermal conductivity measurements, and calculate thermal diffusivity and thermal inertia for eight shergottite (Martian) meteorites: Los Angeles, Northwest Africa [NWA] 7397, NWA 12262, NWA 12269, NWA 12594, NWA 13187, NWA 13366 and NWA 14713, over the range $5 \leq T \leq 300$ K. Measurements follow the procedures as described in [3] utilizing a Quantum Design Physical Property Measurement System (QD-PPMS) at Boston College.

From these data a simple "average" model is proposed for the thermophysical properties of shergottite meteorites. Our efforts inform models of the Martian surface [1,4] and may aid future exploration of the red planet.

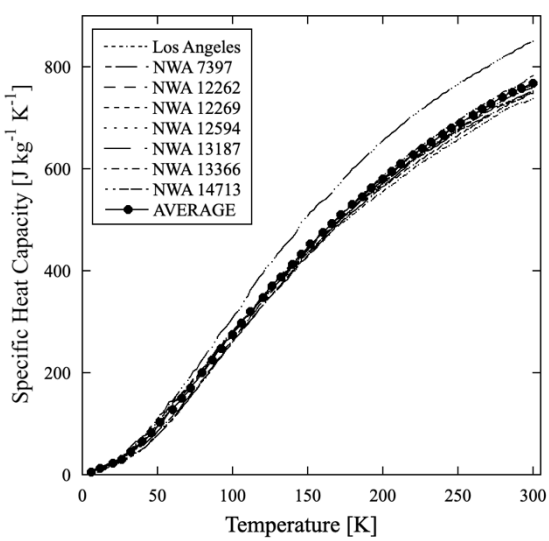

Figure 1: c_p vs. T data are shown for eight shergottite (Martian) meteorites. Error bars are suppressed for plot clarity. Averaged values of c_p vs. T data are represented by filled black circles.

Specific Heat Capacity: Specific heat capacity (c_p) is an intrinsic material property signifying the amount of heat energy necessary to raise the temperature of a certain mass of a material. Measurements of specific heat capacity (c_p vs. T) of the Martian shergottites are shown in Fig. 1. For all samples the temperature dependence is very pronounced, particularly at temperatures below < 100 K and indicate a high-temperature saturation behavior at T > 250 K. Our data are consistent with the analysis of Martian-relevant minerals. [5, 6] The averaged values of c_p varies by a factor of 61.1 between 5 to 100 K and by a factor of 2.8 from 100 to 300 K.

The c_p vs. T data are similar across T with the exception of NWA 7397. Such behavior likely follows the high peak shock-pressure and temperature history experienced by the sample.[7]

Thermal Conductivity: Data for thermal conductivity (κ vs. T) are shown in Fig. 2. Unlike c_p, the value of κ is directly linked to sample porosity, density, fractures, and disorder present in the meteorites. Thus our shergottite data are significantly lower than the literature values of their constituent minerals. The strong

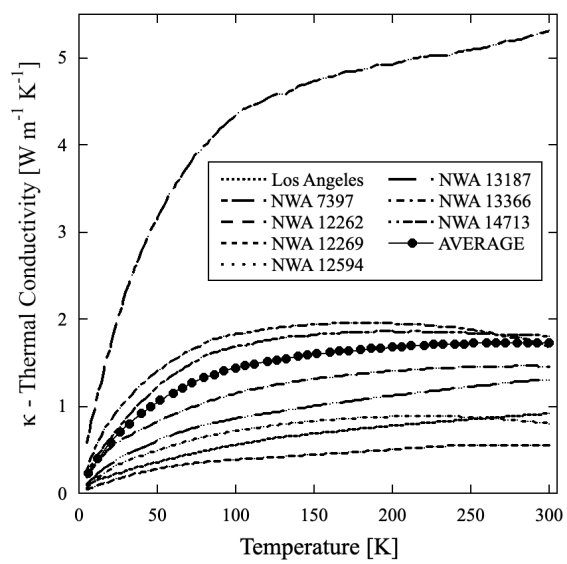

Figure 2: κ vs. T data are shown for eight shergottite (Martian) meteorites. Error bars are suppressed for plot clarity. Averaged values of κ vs. T data are represented by filled black circles.

difference in conductivity of NWA 7397 compared to other samples may reflect a significant difference in its physical state, perhaps related to its shock history. [7]

Our κ vs. T shergottite data reveal that temperature dependence is strong from 5 to 100 K. In contrast, only

slight temperature dependence is indicated from 100 to 300 K. Such curves are typical for Si based, disordered crystalline minerals containing few metallic (Fe, Ni) inclusions. No sharp peaks or evidence of phase transitions are indicated by the κ vs. T data.

The average κ value (filled black circles) changes by a factor of 7.5 from 5-100 K and only changes by a factor of 1.2 from 100 - 300 K.

Thermal Inertia: Thermal inertia (Γ) vs. T values are calculated as ($\Gamma = \rho^{1/2} c_p^{1/2} \kappa^{1/2}$) from ρ (bulk density), c_p, and κ. Data are shown in Fig. 3. Because Γ varies as the square root of the product of heat capacity and thermal conductivity, it is a very strong function of temperature over the entire thermal range in this study.

Our average value of κ (filled black circles) increases by a factor of 21.5 from 5 K to 100 K, and by another factor of 1.9 from 100 K to 300 K. At extremely low temperatures, both conductivity and heat capacity trend toward zero. Since Γ trends continuously toward zero as it approaches 0 K, the data reveals a more well-behaved function of temperature than thermal diffusivity.

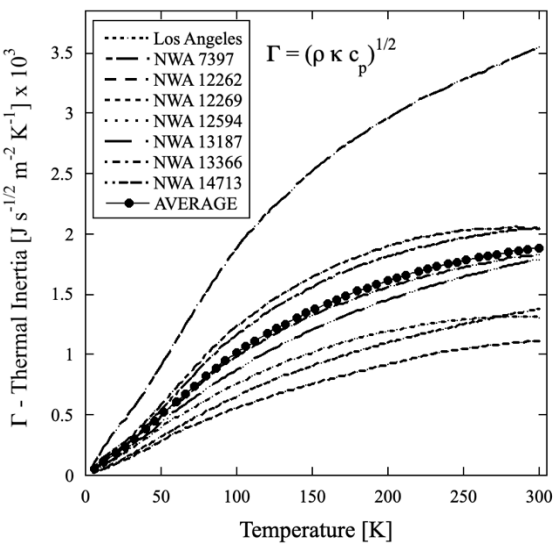

Figure 3: Calculated Γ vs. T data are shown for eight shergottite (Martian) meteorites. Error bars are suppressed. Average values of Γ vs. T are represented by filled black circles.

Thermal Diffusivity: Thermal diffusivity (D_T) vs. T values are calculated as ($D_T = \rho c_p^{-1} \kappa^{-1}$) from ρ (bulk density), c_p, and κ. Data are shown in Fig. 4.

Thermal diffusivity has units of area per unit time and measures how quickly a material transports heat or how quickly temperature can rise. Because thermal inertia and thermal diffusivity are both calculated from heat capacity and thermal conductivity (and density, assumed here to be independent of temperature), both of these properties likewise are strong functions of temperature. In the case of thermal diffusivity, which varies with the ratio of these factors, the temperature dependence flattens out above 100 K. From 100 to 300 K, our average value of D_T (filled black circles) changes by a factor of 2.3. Below 100 K, however, thermal

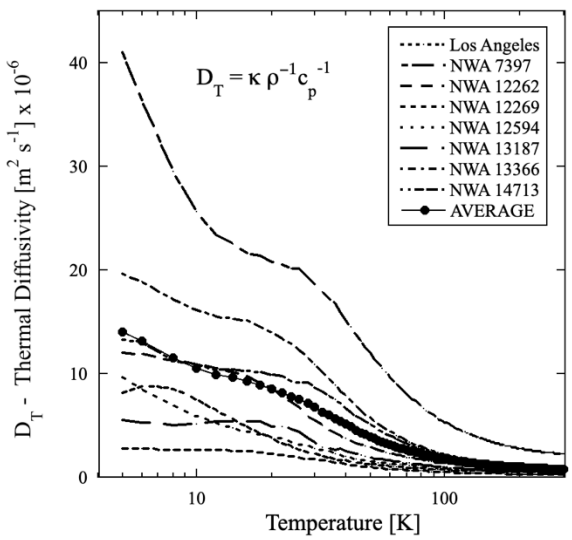

Figure 4: Calculated D_T vs. T data are shown for eight shergottite (Martian) meteorites. Error bars are suppressed. Average values of D_T vs. T are represented by filled black circles.

conductivity, κ, and specific heat capacity, c_p, exhibit vastly different behaviors with temperature: the denominator (specific heat capacity) trends towards zero faster than thermal conductivity. Therefore, as temperature drops the thermal diffusivity can increase to a maximum, before falling toward zero at absolute zero. From 5-100 K, our average value of D_T changes by a factor of 8.0.

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